

## Question ID: cb29c54c

### Question

A physics class is planning an experiment about a toy rocket. The equation  $y = -16(x - 5.6)^2 + 502$  gives the estimated height  $y$ , in feet, of the toy rocket  $x$  seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the  $xy$ -plane?

### Answer

- A. This toy rocket reaches an estimated maximum height of **502** feet **16** seconds after it is launched into the air.
- B. This toy rocket reaches an estimated maximum height of **502** feet **5.6** seconds after it is launched into the air.
- C. This toy rocket reaches an estimated maximum height of **16** feet **502** seconds after it is launched into the air.
- D. This toy rocket reaches an estimated maximum height of **5.6** feet **502** seconds after it is launched into the air.